

Introduction and Overview

Digital Image Processing

ECE 253

ECE 253: Digital Image Processing (Syllabus)

- Tuesdays/Thursdays: 5-6:20pm, on zoom <https://ucsd.zoom.us/j/8314692956>
- Instructor: Prof. Nguyen. Office Hours: T/Th 3:30-4:30pm, on zoom
- TA: Huong Hoang. Office Hours:
 - Mondays, 3:30p-4:30pm, <https://ucsd.zoom.us/j/95651285075>
 - Wednesdays, 1:30pm-2:30pm, : <https://ucsd.zoom.us/j/96535731549>
 - Fridays, 1:30pm-2:30pm. FAH Collab Space 2301
- Podcast, Piazza, Canvas
- Course Requirement & Grading
 - Project: Team of 2 to 3 students
 - Project Proposal (11/6) 25% Submit by 11:59pm
 - Project Update Presentation (11/24, 11/25 11/26) 25% (7-10pm on zoom)
 - Final Project Presentation (12/10, 12/11, 12/12) 25% (7-10pm on zoom)
 - Project Report (12/14) 25% Submit by 11:59pm
 - Zoom link for presentation: <https://ucsd.zoom.us/j/8314692956>

ECE 253: Digital Image Processing (Syllabus)

- **Course grade:**
 - A+: 97% - 100%
 - A: 90% - 96.9%
 - A-: 87% - 89.9%
 - B+: 83% - 86.9%
 - B: 77% - 82.9%
 - B-: 74% - 76.9%
 - Passing grade for comprehensive exam: B or better
- **Recommended Textbook:**
 - *Image Processing for Engineers* by Andrew E. Yagle and Fawwaz T. Ulaby. Download free copy at <https://ip.eecs.umich.edu/>
 - *Digital Image Processing* by Rafael C. Gonzalez and Richard E. Woods, 4th edition, Pearson.
- **Recommended Preparation:** ECE 109, ECE 153, ECE 161A

ECE 253: Digital Image Processing (Syllabus)

Topics

- Overview of Digital Image Processing
- Review of DSP: Linear-Phase filter, Filter Design, DFT, etc.
- Digital Camera: Image Sensors & Camera Processing Pipeline
- 2D Image & System
- Image Enhancement: Histogram Equalization, Edge Sharpening, Median Filtering, Directional Filtering
- Color Image Processing
- Image Interpolation & Decimation, Multirate DSP
- Wavelets & Multiresolution Analysis
- Image Restoration: Denoising, DeFogging/DeHazing, De-Turbulence
- Image Compression: JPEG & JPEG2000

Collaboration Policy

- Final project:
 - You can use public-domain code on GitHub (posted by paper’s authors). You must provide appropriate reference for the source code.
 - You must clearly indicate the modification/addition to existing code.
 - **Project Proposal: due 11:59pm on 11/6.** Come to office hour to discuss project ideas. Set up appointment if needed.
 - More on Final Project later.
- Ask and answer questions on Piazza, not email
- Post **publicly** (optionally anonymously)
 - Conceptual questions and high-level questions about assignments & class materials.
- All other posts must be **private** to “Instructors” (includes instructor and instructional assistants)
 - Questions related to your own project
- Piazza is the official, University-sanctioned discussion forum
 - **Do not use Piazza to solicit others to an alternative forum**

Academic Integrity & Student Conduct Policy

- Integrity of scholarship is essential for an academic community. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.
- If you commit a violation, then you will be reported to the Academic Integrity Office for violating UCSD's Policy on Integrity of Scholarship.
- Maintaining an academic community free from disruption/obstruction, physical abuse, harassment, and other conduct inconsistent with UCSD's Principles of Community is each of our responsibility and essential to promoting mutual respect and academic rigor. Non-academic student misconduct will be reported to the Center for Student Accountability, Growth, and Education for violating the University Standards of Conduct.

History of Digital Camera

- **Steven Sasson** (1969). First Digital Camera.
- **George Smith and Willard Boyle** at AT&T's Bell Labs conceived the charge-coupled device (CCD), a semiconductor memory concept that allowed easy manipulation of charges in silicon structures.
- **Michael Tompsett** (AT&T), invented the CCD image sensor that first demonstrated the electronic photography and video in use today.
- The first self-contained digital camera (1975) is credited to an engineer at Kodak.
- **Eric Fossum** (1993), invented the first CMOS active sensor camera-on-a-chip (JPL).



Steven Sasson



Eric Fossum

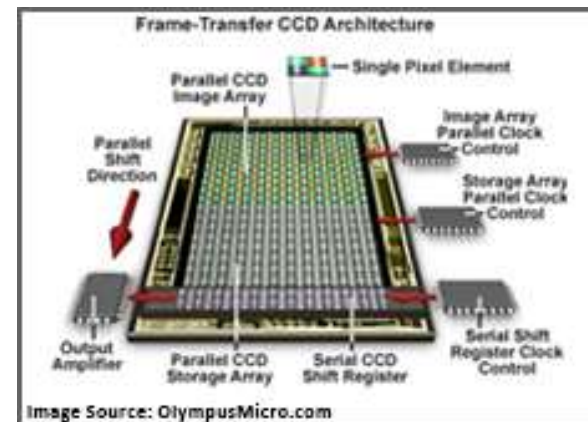
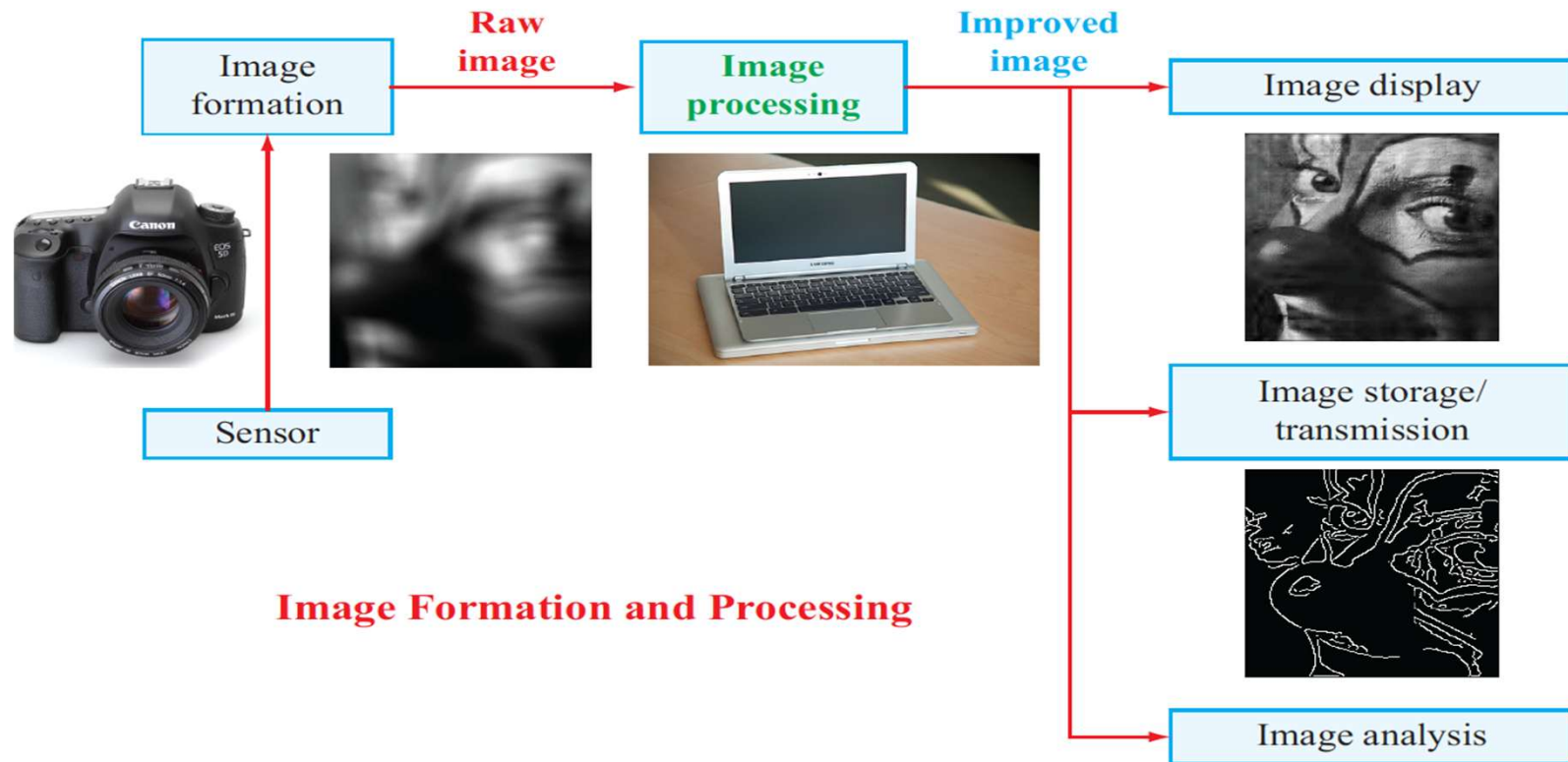


Image Formation and Processing



After an image is formed by a sensor, image processing tools are applied for many purposes, including changing its scale and orientation, improving its information content, or reducing its digital size.

Electromagnetic Spectrum

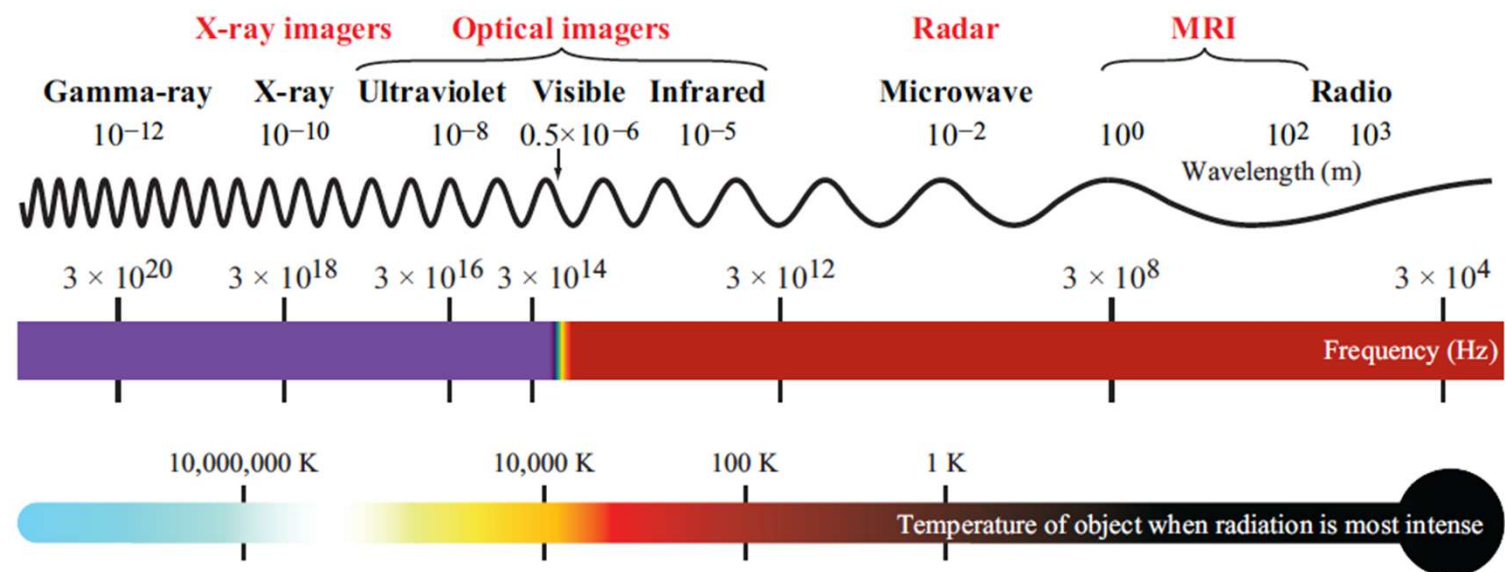


Figure 1-2 Electromagnetic spectrum.

Digital Cameras

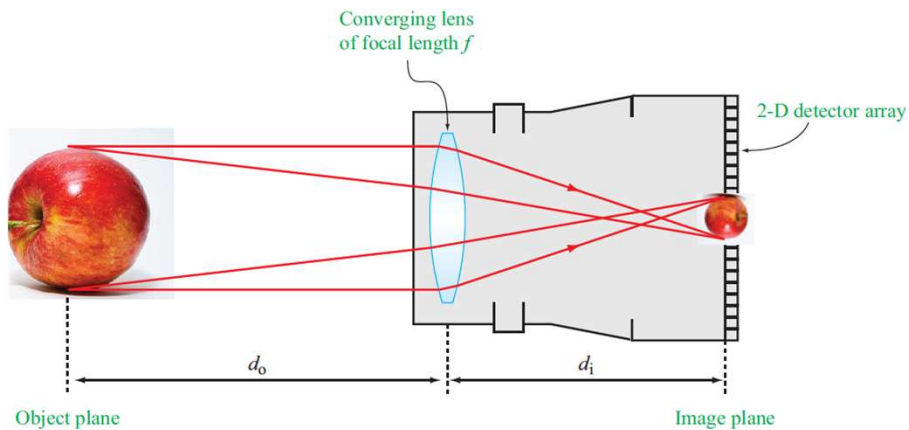


Figure 1-3 Camera imaging system.

$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f} \quad (\text{lens law})$$

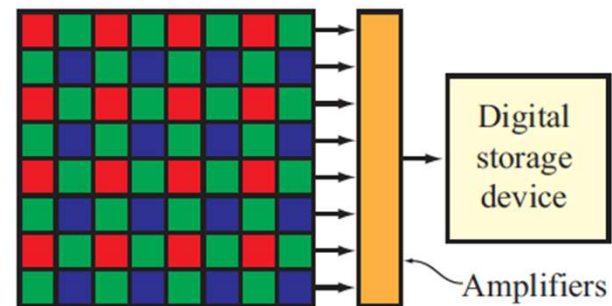
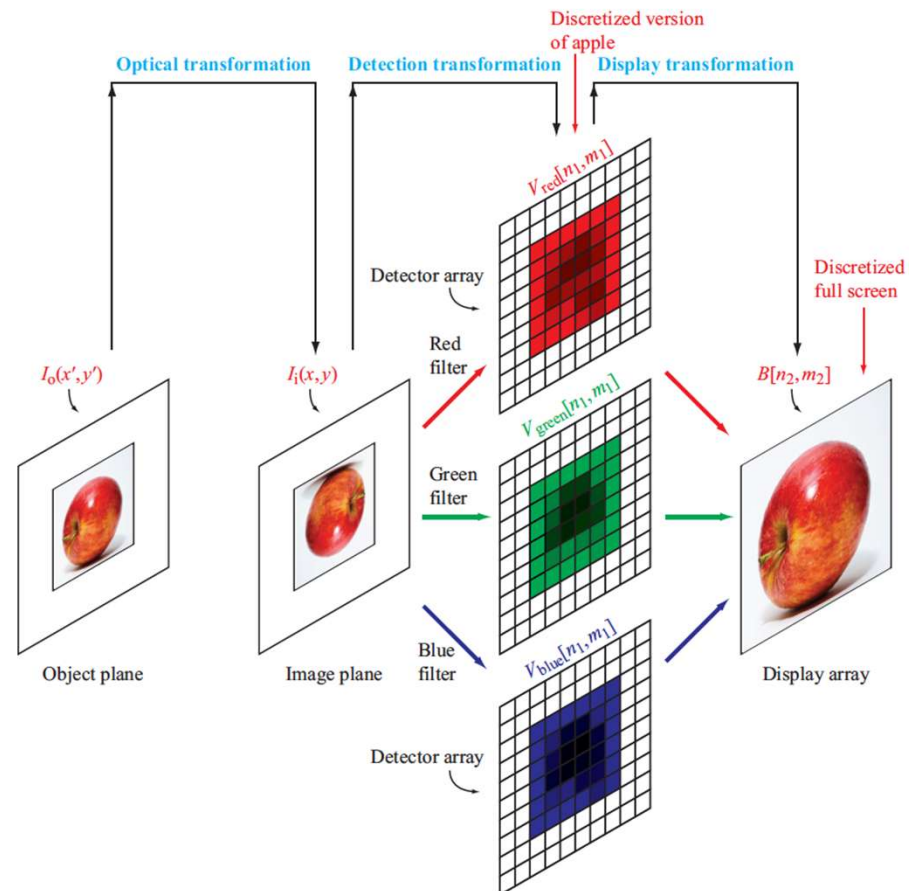
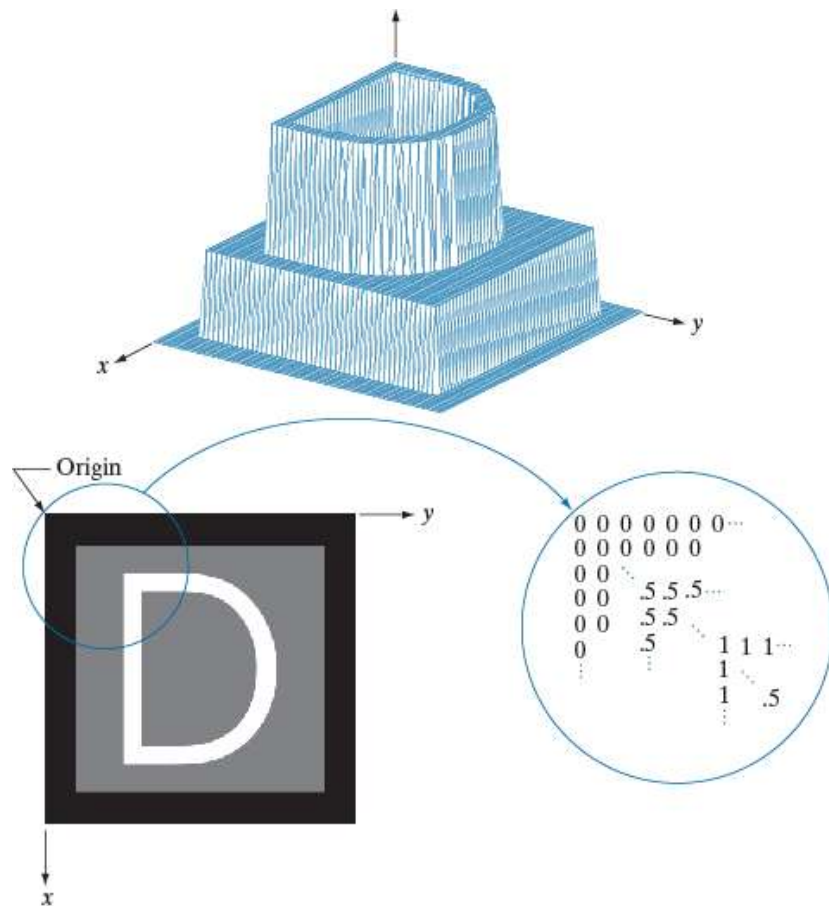


Figure 1-4 An active pixel sensor uses a 2-D array of photodetectors, usually made of CMOS, to detect incident light in the red, green, and blue bands.

Image Detection and Display



Representing an image



A two-dimensional function $f(x,y)$, where x and y are spatial coordinates

The amplitude of f at the coordinates (x,y) is called the intensity or gray level at that point

A *digital* image is composed of a finite number of elements at discrete points

The elements are called picture elements (pixels, pels) or image elements

Examples

- Gamma-ray imaging
- X-ray imaging
- Ultraviolet imaging
- Visible light imaging
- Infrared imaging
- Microwave imaging
- Radio imaging

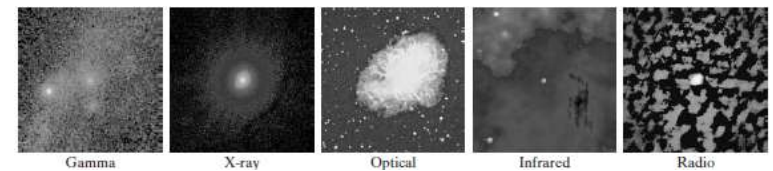
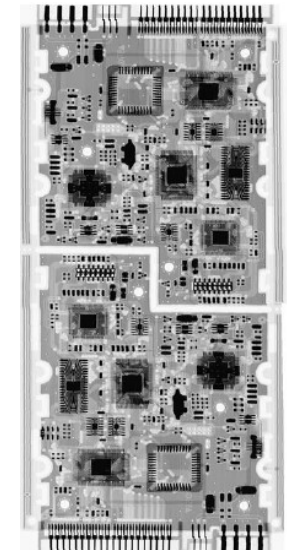
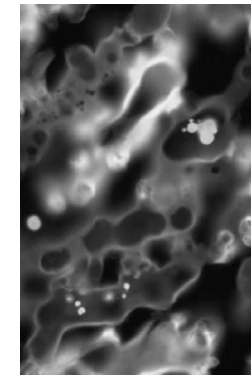
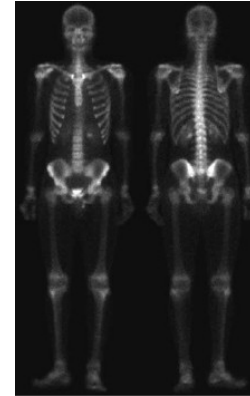
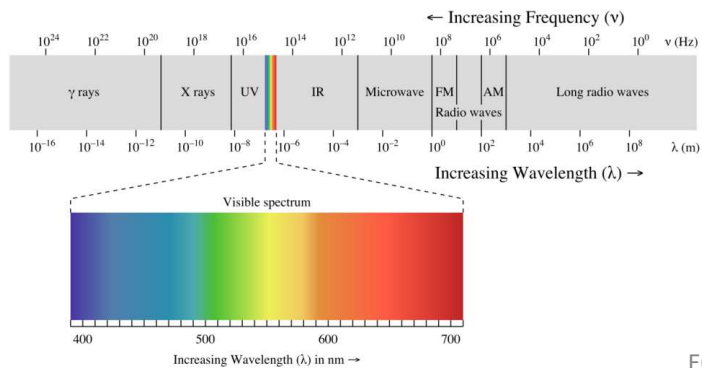


Image Processing vs. Computer Vision

- **Image Processing**
 - Manipulate raw images to enhance features: Noise reduction, Color Correction, etc.
 - Prepare them for image/video analysis
- **Computer Vision**
 - Interpret and understand content of image/video
 - Object recognition, image segmentation, scene understanding, etc.

Image Processing



Stutterstock



Yang et. al. ACCV 2022



Stutterstock

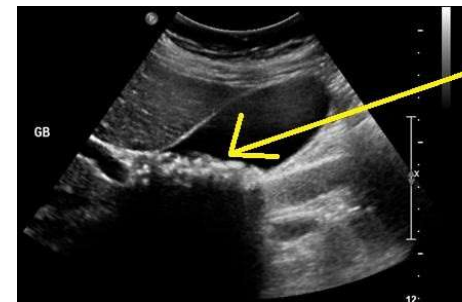


<https://www.analyticsvidhya.com/blog/2018/04/solving-an-image-captioning-task-using-deep-learning/>



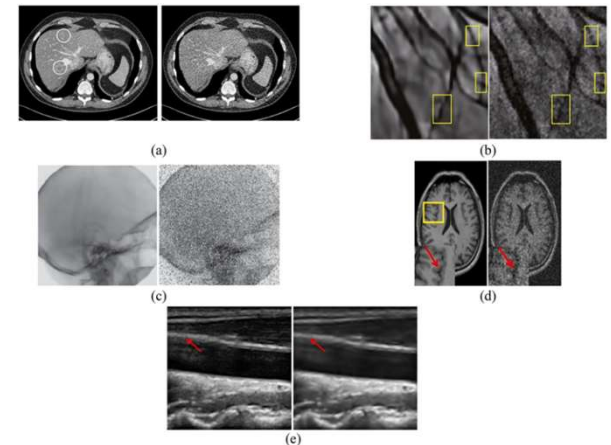
Pinterest

ECE 253, Fall 2025



Gallstones noted near the neck of the gallbladder

<https://www.harmonicmedicalsonography.com/2022/11/16/diagnosing-gallstones-ultrasound/>



a. CT; b. angiography; c. angiography; d. MRI; e. human carotid artery.
https://www.researchgate.net/figure/Effect-of-different-noise-types-on-medical-images-anormal-dose-and-quarter-dose-abdomen_fig1_362478239

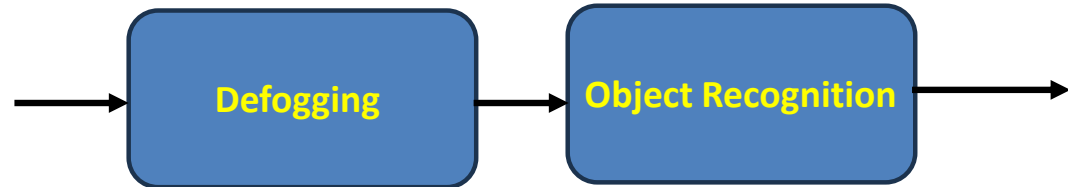
Image Processing



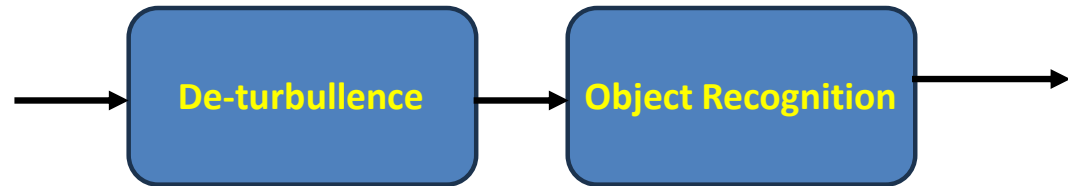
Stutterstock



Yang et. al. ACCV 2022



Stutterstock



ECE 253 – Final Project

- Demonstrate the usage of Digital Image Processing.
- Free to use other works (Deep learning, etc.) in image and video applications: Object recognition, semantic segmentation, scene understanding, etc.
- Need to collect your own data (for fine-tuning).
- Comparing with and without Digital Image Processing.
- Discussion of Pros & Cons of various Digital Image Processing methods.

Digital image processing

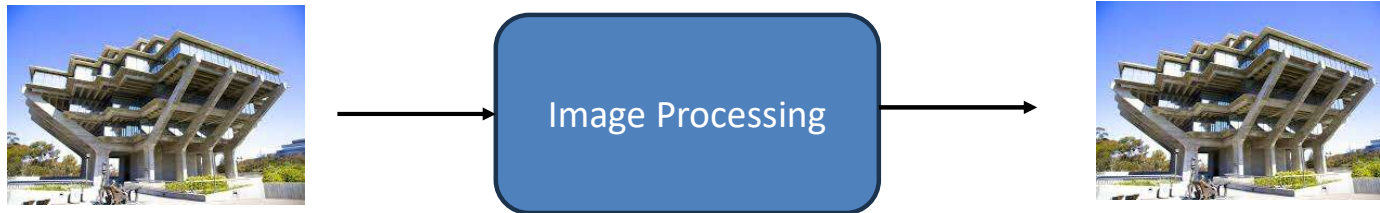


Image Enhancement

Histogram equalization, edge sharpening, median filtering, etc.

Image Interpolation & Decimation

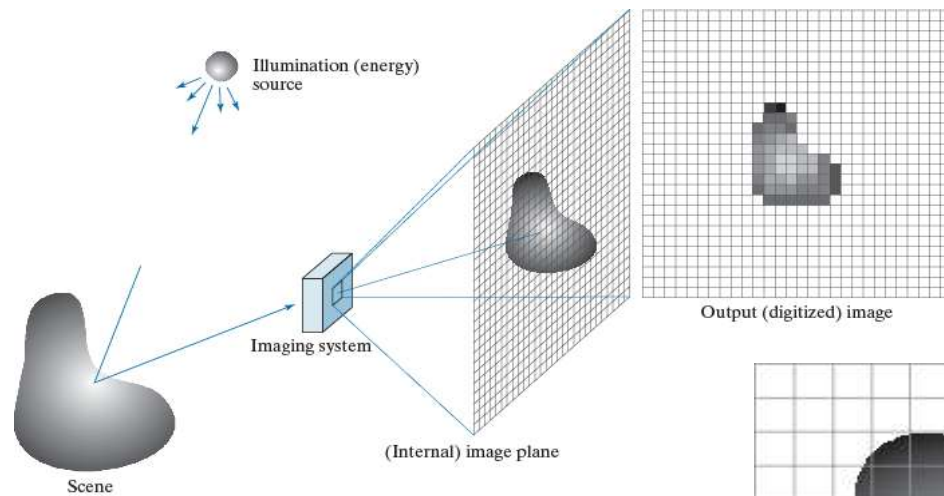
Image Restoration

Denoising, Defogging, Dehazing, De-Turbulence

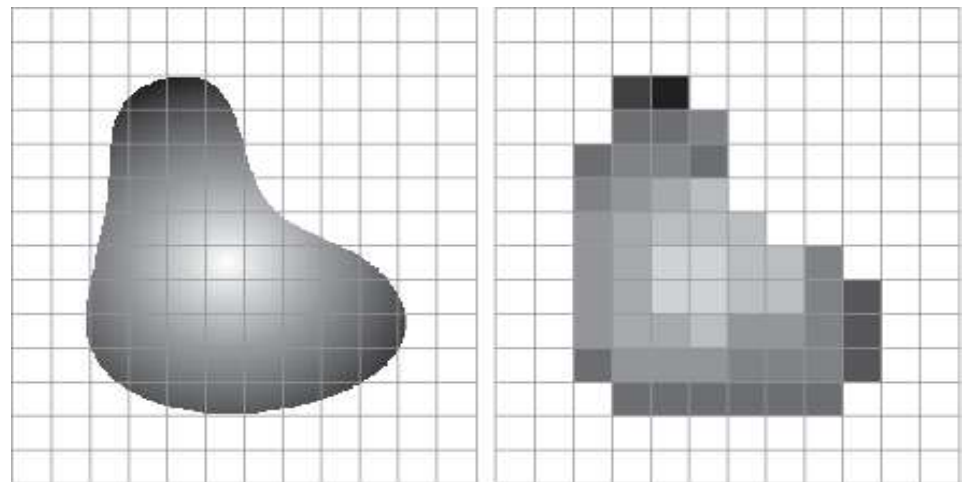
Image Compression

JPEG, JPEG2000

Image acquisition

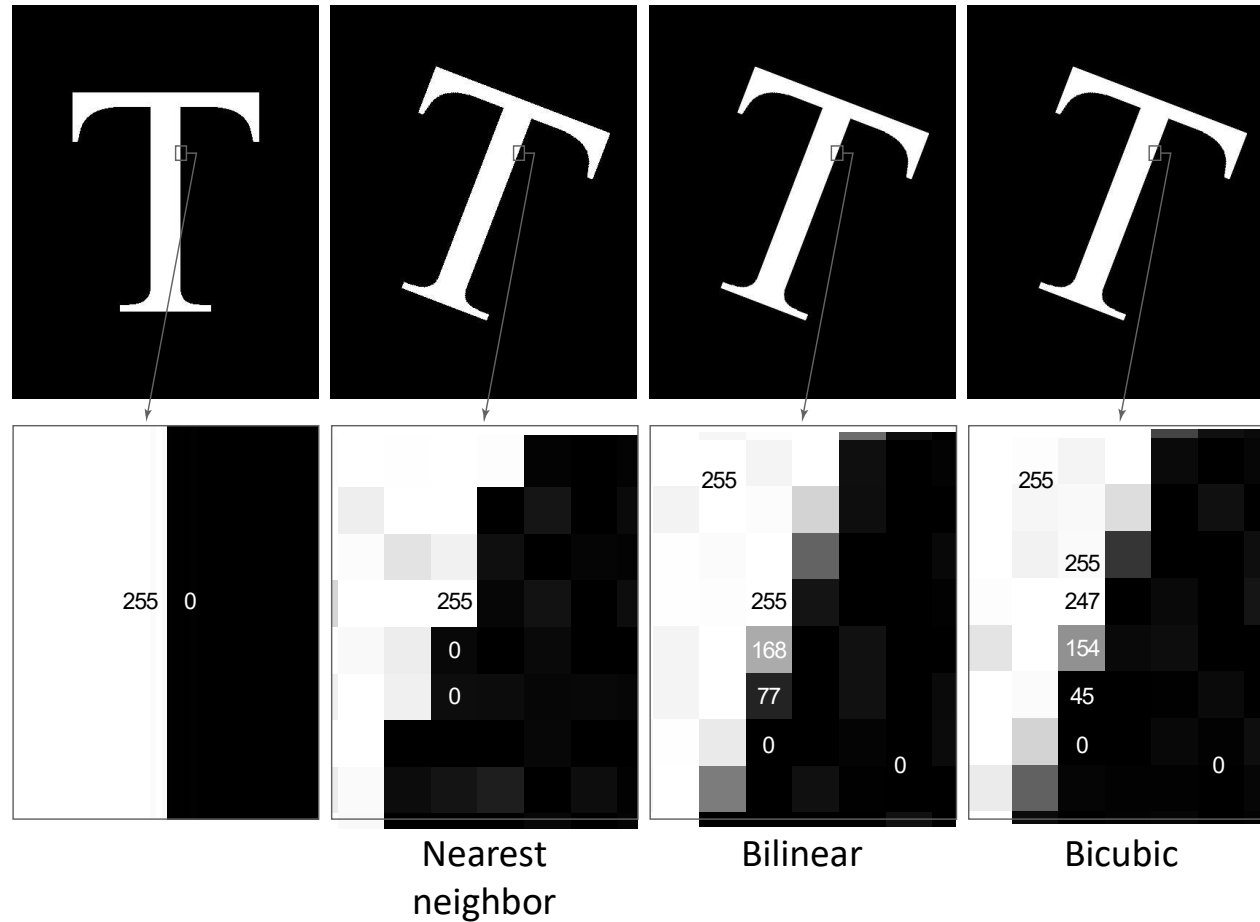


8064 x 6048 px, 48 MP (Iphone 14 & 15 Pro)
48MP → Effective 12MP
8 bits vs. 16 bits
8-12 colors



Sampling and quantization

Geometric transformation & Interpolation



Filtering and Enhancement

- Intensity transformations
- Spatial filtering



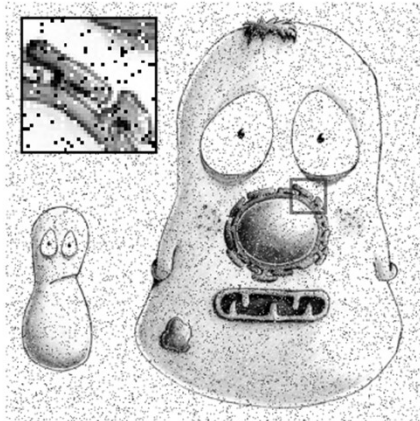
Low-pass filter

Gamma
correction

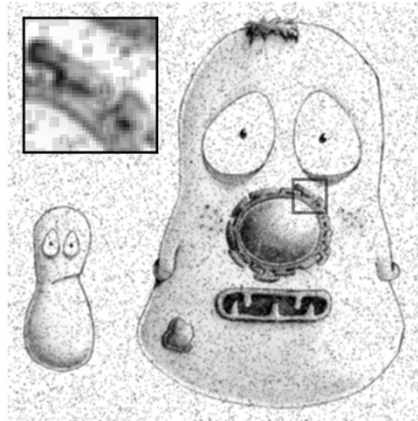


Filtering and Enhancement

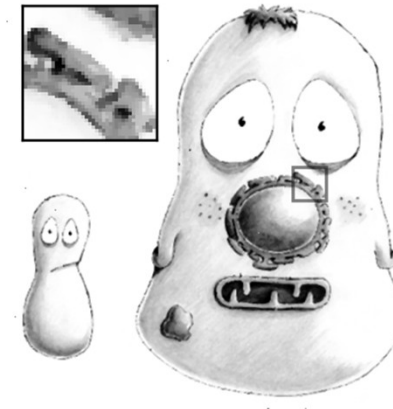
(A) Salt & pepper noise



(B) Mean filter



(C) Median filter



(A) Horizontal gradient



(B) Vertical gradient



(C) Gradient magnitude



Filtering and Enhancement

- Noise models
- Noise reduction, Denoising

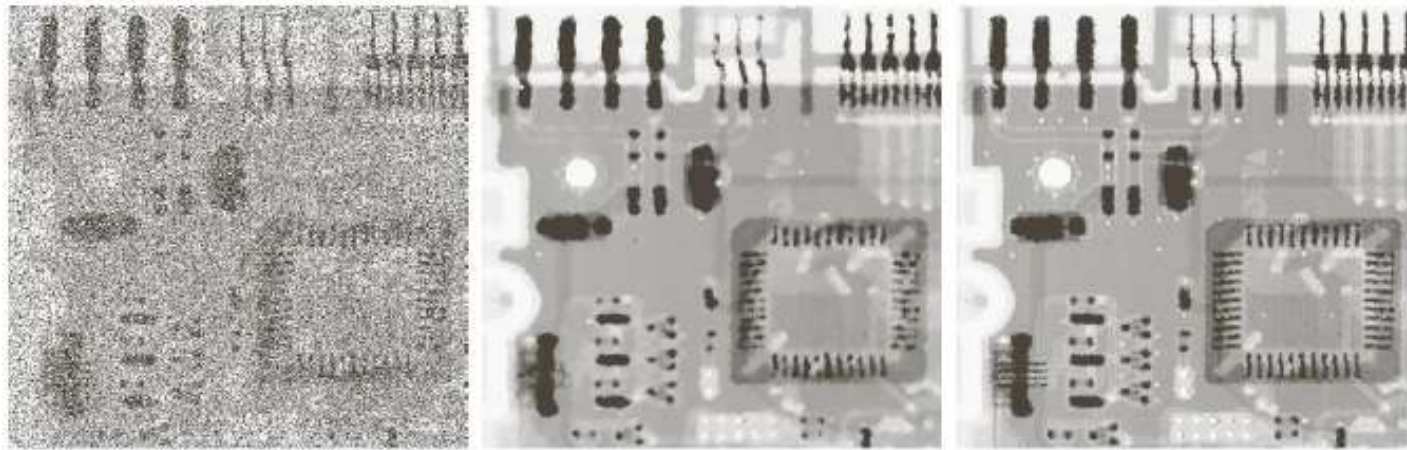
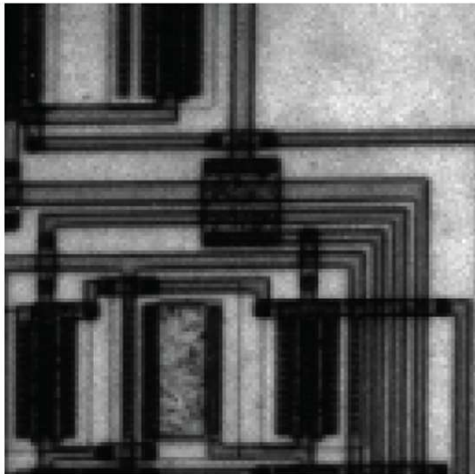
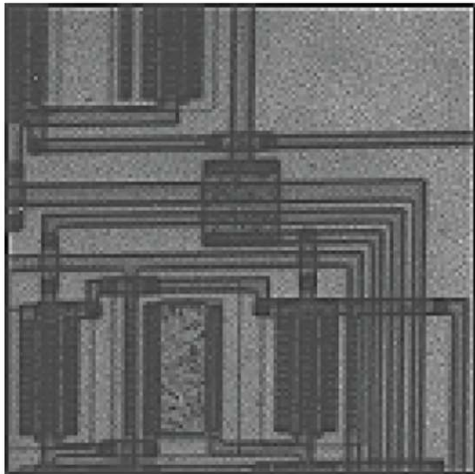


Image Sharpening

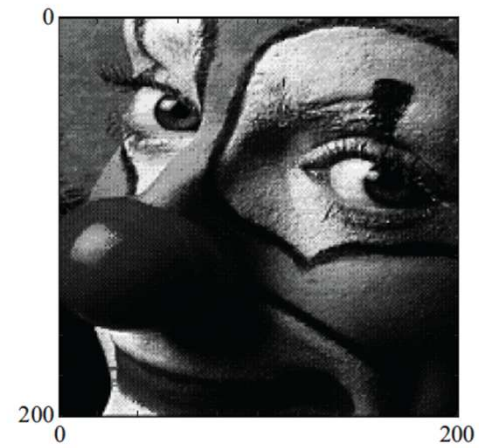


(a) Original image

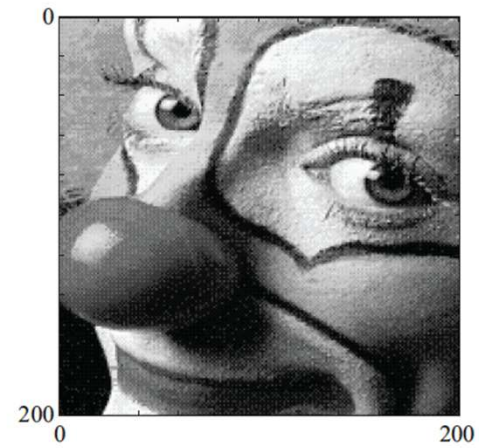


(b) Sharpened image

Image Brightening by Histogram Equalization

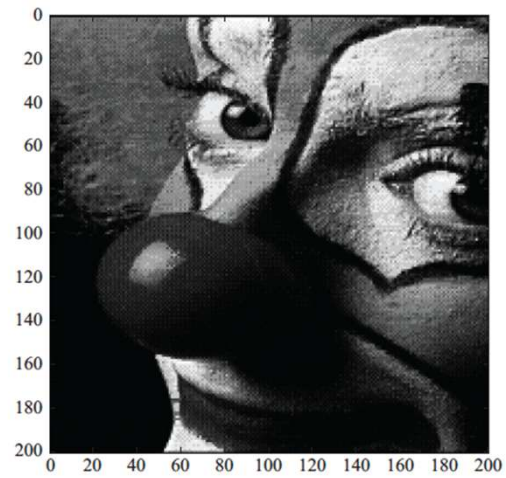


(a) Dark clown image

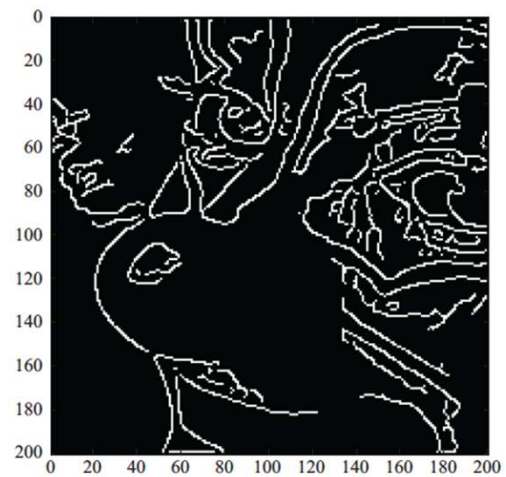


(b) Brightened clown image

Edge Detection

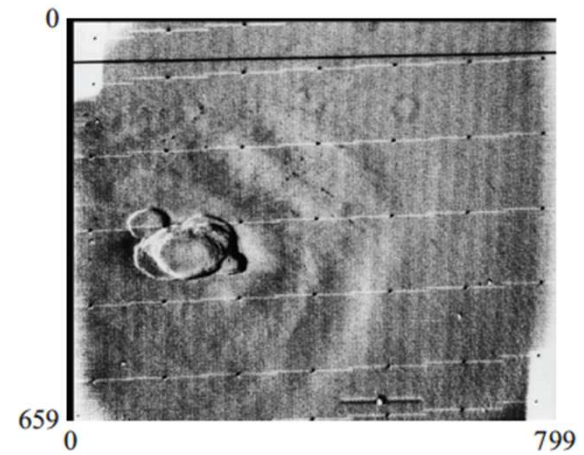


(a) Original clown image

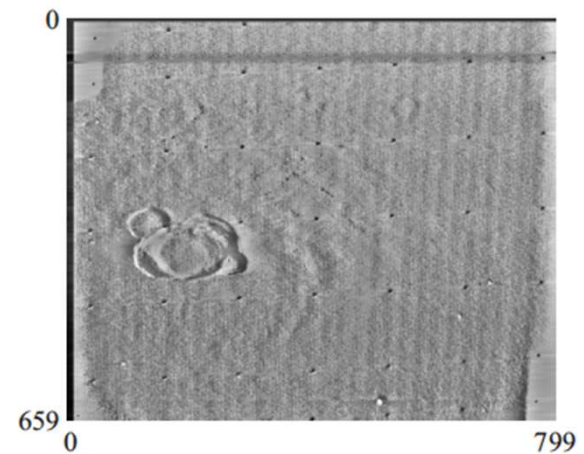


(b) Edge-detected image

Notch Filtering

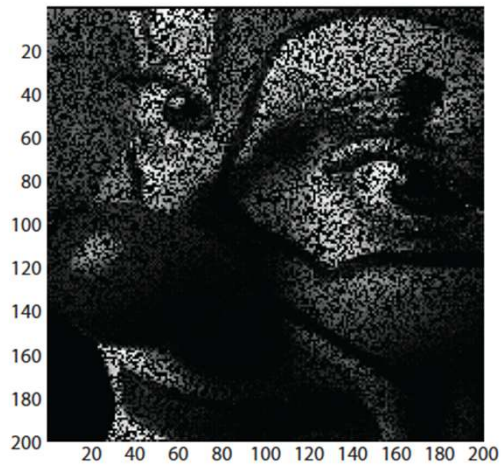


(a) Original Mariner image

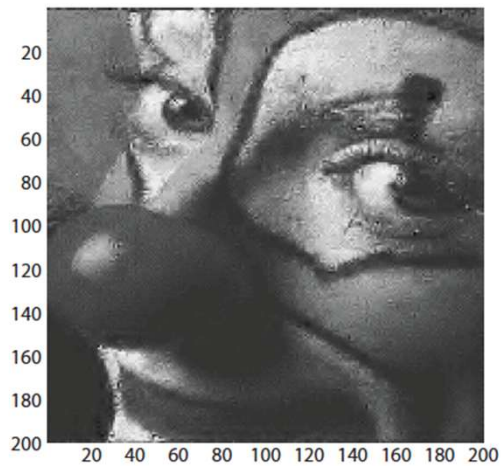


(b) Notch-filtered Mariner image

Image Inpainting



(a) Image with missing pixels



(b) Inpainted image

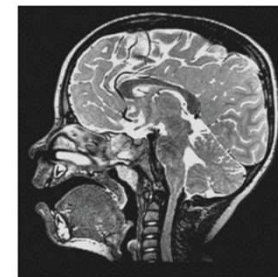
Image Restoration by Deconvolution



(a) Unfocused MRI image



(b) Deterministically refocused MRI image



(c) Stochastically refocused MRI image

Image Segmentation



(a) Noisy image



(b) Segmented image

Motion Deblurring



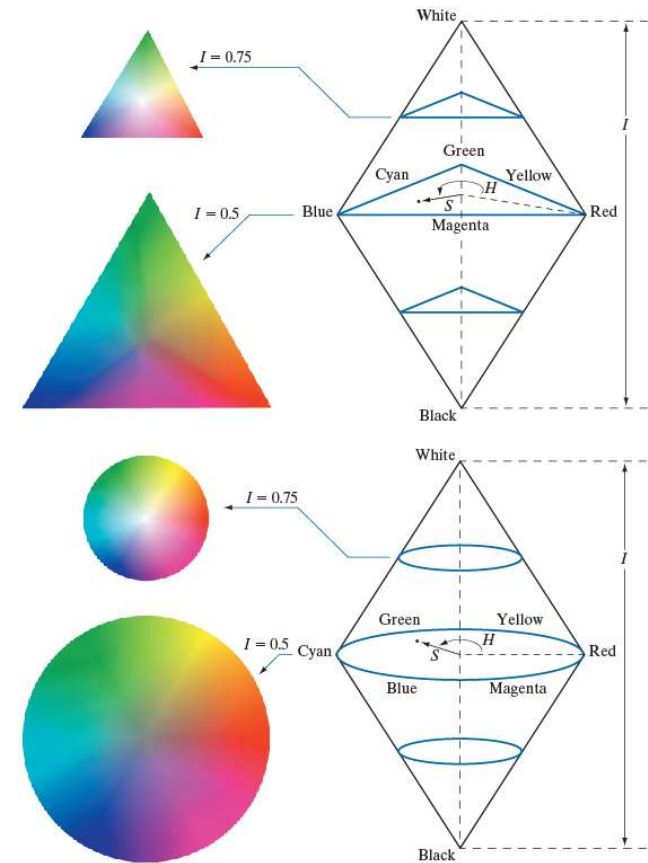
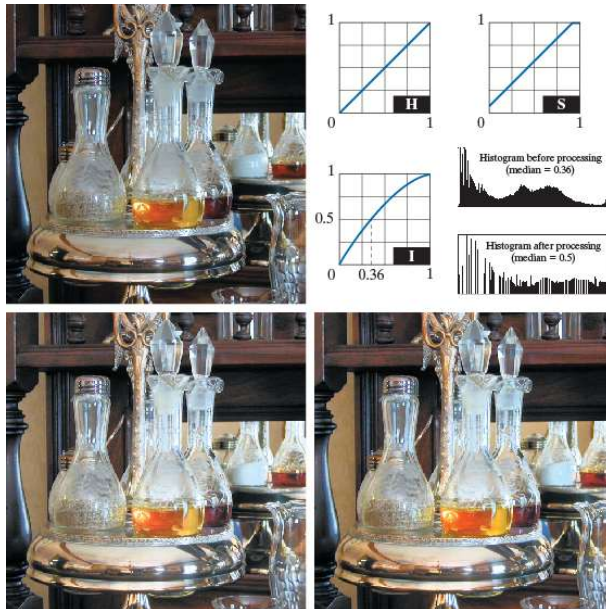
(a) Motion-blurred Christmas tree



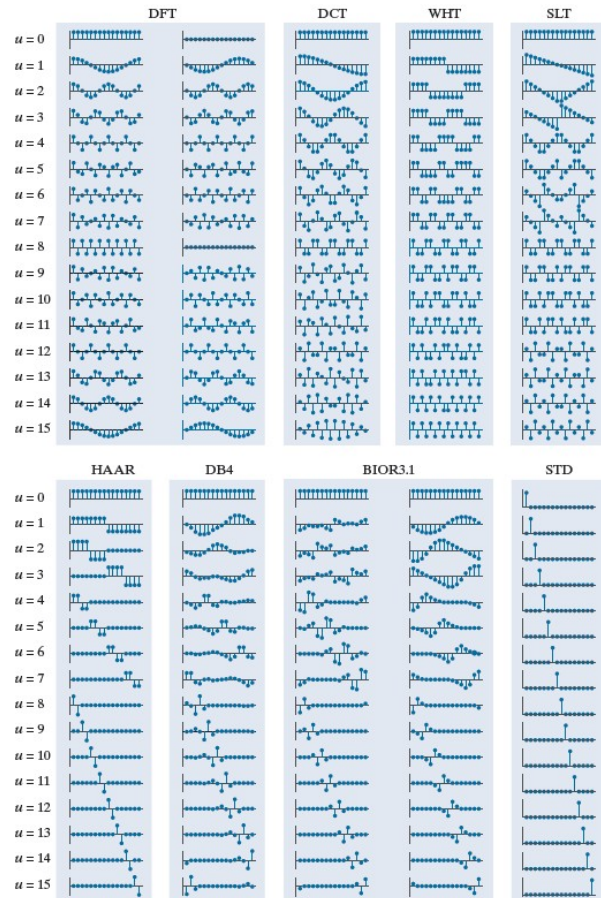
(b) Deblurred Christmas tree

Color Image Processing

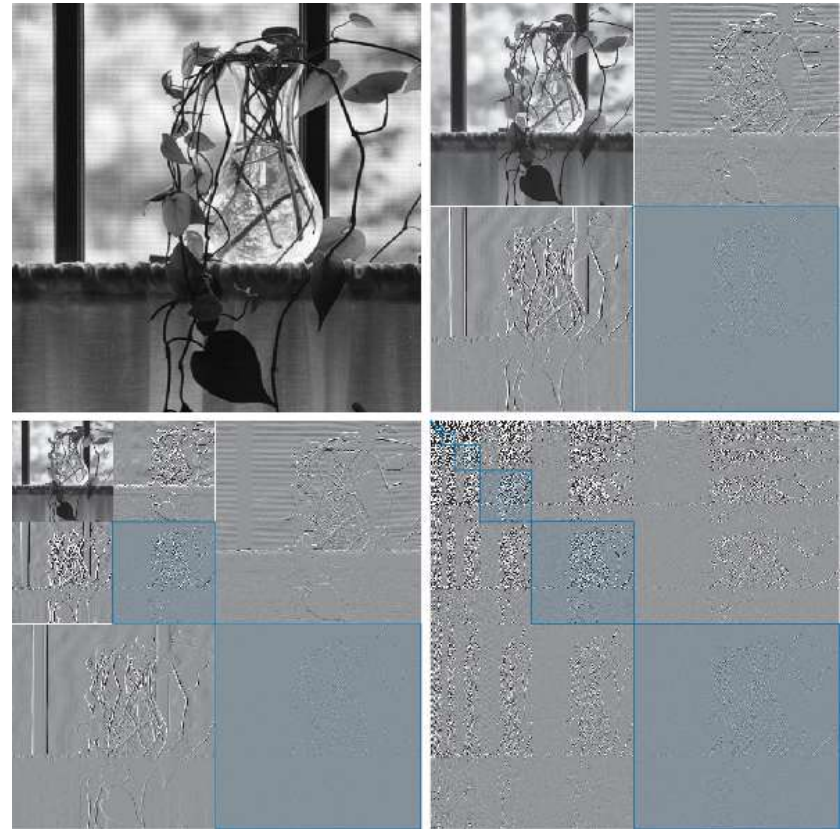
- Color models
- Color transformations



Wavelets and Multiresolution Analysis

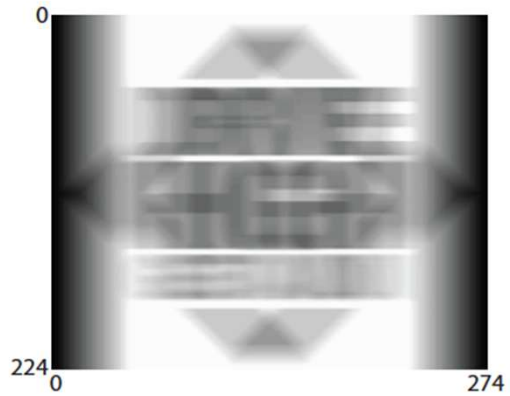


Basis vectors

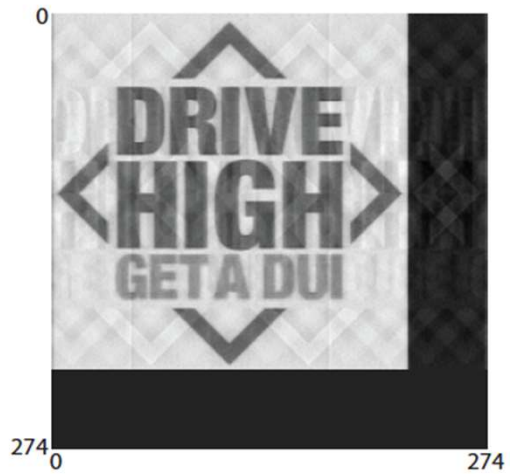


Wavelet and Haar transform

Motion-Blur Deconvolution

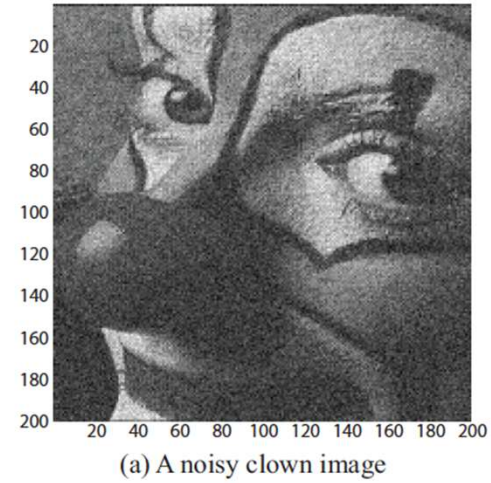


(a) Original motion-blurred image caused by taking a photograph in a moving car

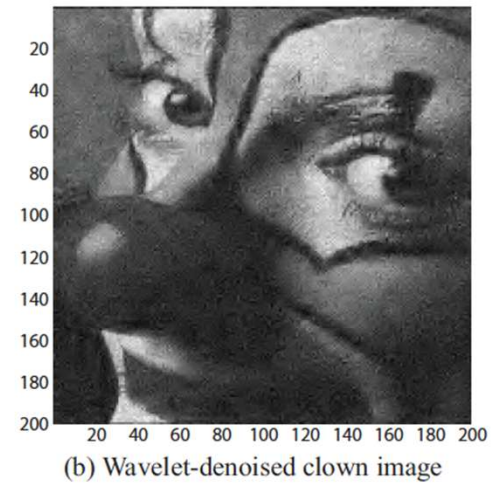


(b) Motion deblurred image

Denoising with Wavelets

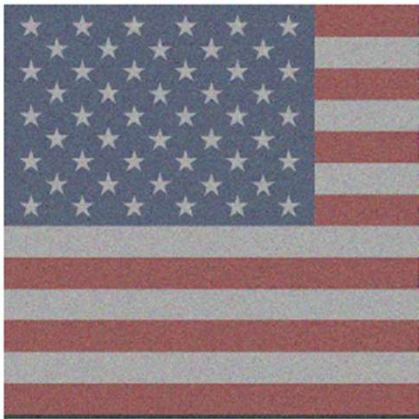


(a) A noisy clown image

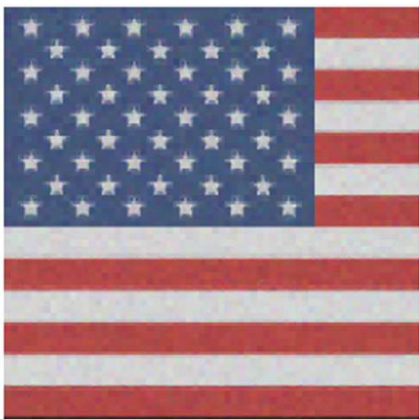


(b) Wavelet-denoised clown image

Wavelet-based Denoising



(a) Noisy flag image



(b) Denoised image

Histogram Equalization



(a) Original dark toucan image



(b) Brightened image

Image compression and watermarking



SMART COMPRESSION:
✓ **QUALITY PRESERVED**

OVERCOMPRESSED:
✗ **IMPACTED QUALITY**



Morphological Image Processing

(A) Erosion



(B) Dilation



(C) Opening



(D) Closing



Image DeHazing/DeFogging

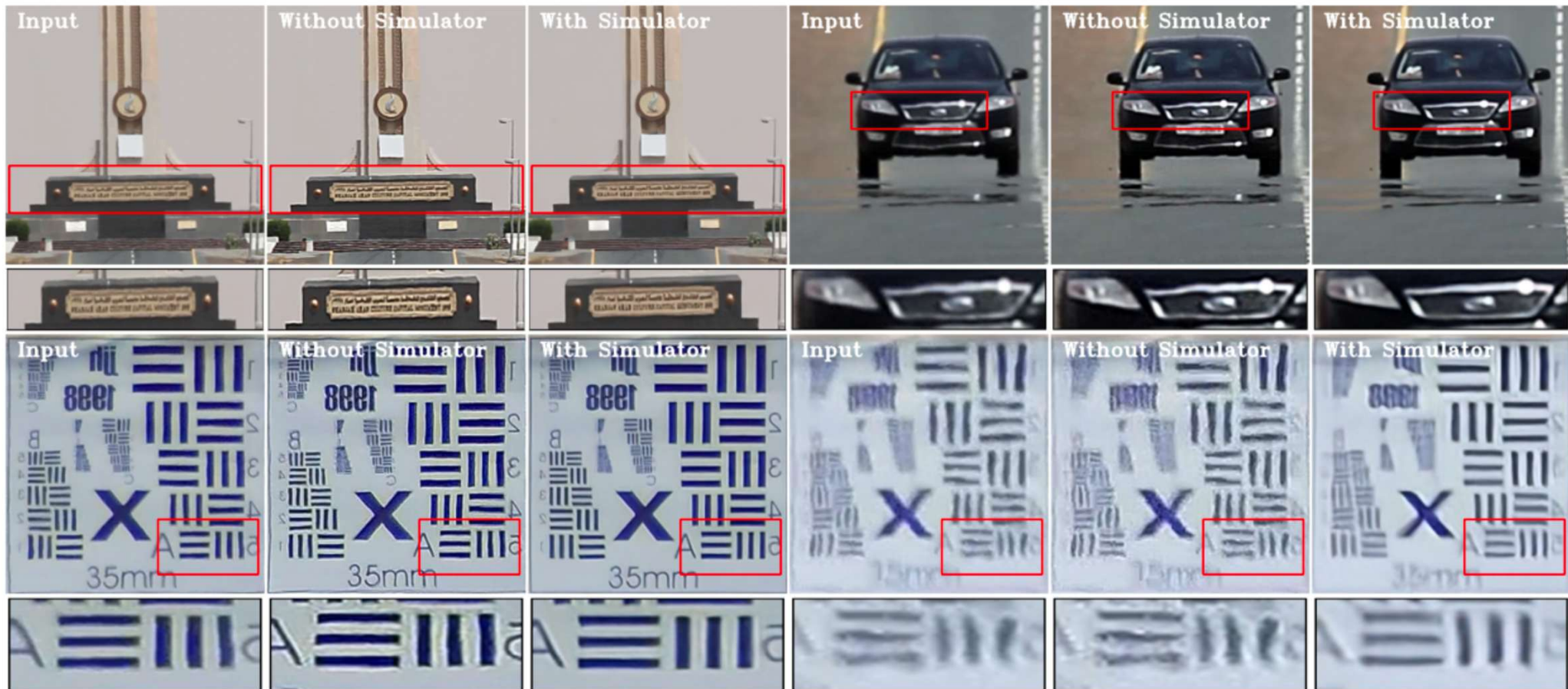


Foggy day



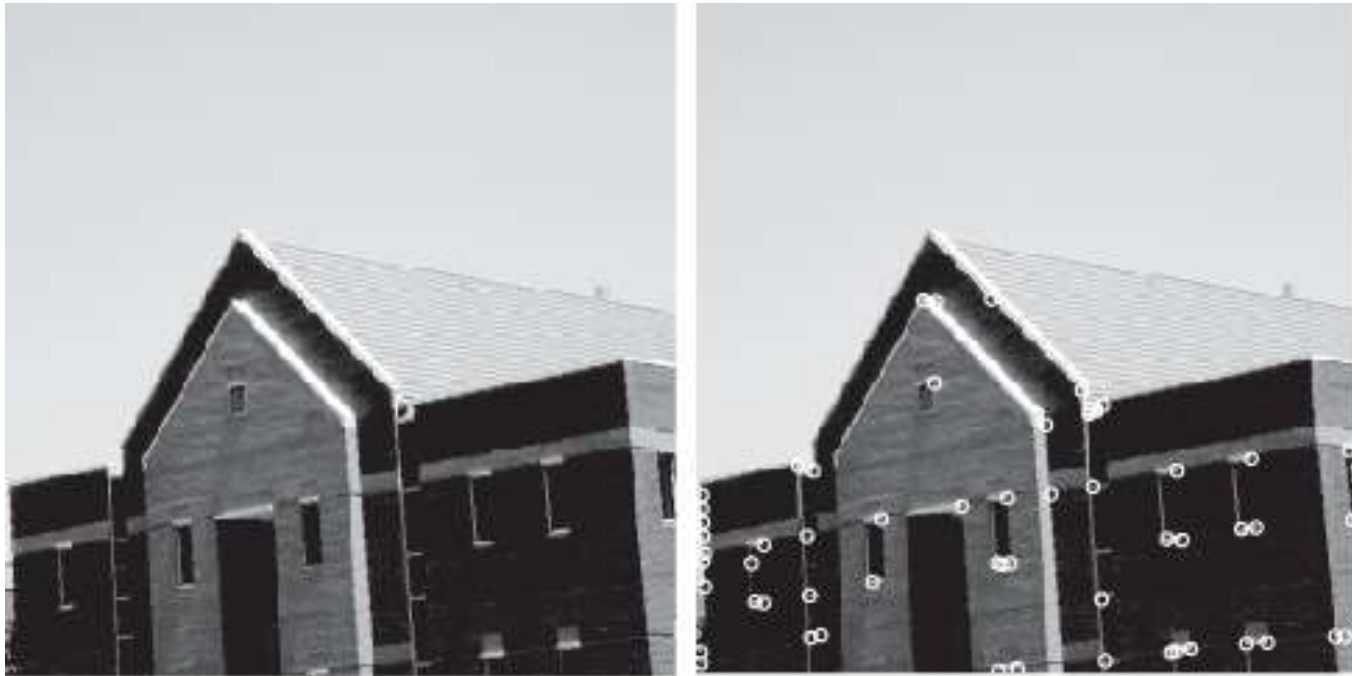
Clear day

Image De-Turbulence



Feature Extraction

- Feature detection



Review of vectors and matrices

- Vectors and matrices
- Vector transpose and matrix transpose
- Vector-vector inner (or dot) product
- Matrix-vector multiplication
- Matrix-matrix multiplication

Review of Vectors and Matrices

Some Definitions

An $m \times n$ (read "m by n") **matrix**, denoted by $\mathbf{A}$, is a rectangular array of entries or elements (numbers, or symbols representing numbers) enclosed typically by square brackets, where m is the number of rows and n the number of columns.

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Review of Vectors and Matrices

Definitions (Con't)

- **A** is *square* if $m = n$.
- **A** is *diagonal* if all off-diagonal elements are 0, and not all diagonal elements are 0.
- **A** is the *identity matrix* (**I**) if it is diagonal and all diagonal elements are 1.
- **A** is the *zero* or *null matrix* (**0**) if all its elements are 0.
- The *trace* of **A** equals the sum of the elements along its main diagonal.
- Two matrices **A** and **B** are *equal* iff they have the same number of rows and columns, and $a_{ij} = b_{ij}$.

Review of Vectors and Matrices

Definitions (Con't)

- The **transpose** $\mathbf{A}^T$ of an $m \times n$ matrix $\mathbf{A}$ is an $n \times m$ matrix obtained by interchanging the rows and columns of $\mathbf{A}$.
- A square matrix for which $\mathbf{A}^T = \mathbf{A}$ is said to be **symmetric**.
- Any matrix $\mathbf{X}$ for which $\mathbf{XA} = \mathbf{I}$ and $\mathbf{AX} = \mathbf{I}$ is called the **inverse** of $\mathbf{A}$.
- Let c be a real or complex number (called a **scalar**). The **scalar multiple** of c and matrix $\mathbf{A}$, denoted $c\mathbf{A}$, is obtained by multiplying every elements of $\mathbf{A}$ by c . If $c = -1$, the scalar multiple is called the **negative** of $\mathbf{A}$.

Review of Vectors and Matrices

Definitions (Con't)

A **column vector** is an $m \times 1$ matrix:

$$\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix}$$

A **row vector** is a $1 \times n$ matrix:

$$\mathbf{b} = [b_1, b_2, \dots, b_n]$$

A column vector can be expressed as a row vector by using the transpose:

$$\mathbf{a}^T = [a_1, a_2, \dots, a_m]$$

Review of Vectors and Matrices

Some Basic Matrix Operations

- The **sum** of two matrices **A** and **B** (of equal dimension), denoted **A** + **B**, is the matrix with elements $a_{ij} + b_{ij}$.
- The **difference** of two matrices, **A** – **B**, has elements $a_{ij} - b_{ij}$.
- The **product**, **AB**, of $m \times n$ matrix **A** and $p \times q$ matrix **B**, is an $m \times q$ matrix **C** whose (i,j) -th element is formed by multiplying the entries across the i th row of **A** times the entries down the j th column of **B**; that is,

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{pj}$$

Review of Vectors and Matrices

Some Basic Matrix Operations (Con't)

The *inner product* (also called *dot product*) of two vectors

$$\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

is defined as

$$\begin{aligned} \mathbf{a}^T \mathbf{b} &= \mathbf{b}^T \mathbf{a} = a_1 b_1 + a_2 b_2 + \cdots + a_m b_m \\ &= \sum_{i=1}^m a_i b_i. \end{aligned}$$

Note that the inner product is a scalar.

Element-wise vs. matrix operations

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

Hadamard product (i.e., elementwise product)

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} & a_{12}b_{12} \\ a_{21}b_{21} & a_{22}b_{22} \end{bmatrix}$$

Matrix product

On the other hand, the *matrix product* is given by

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}$$